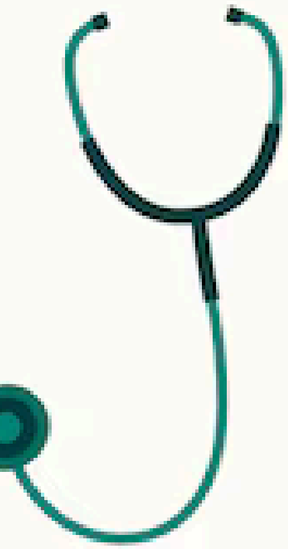


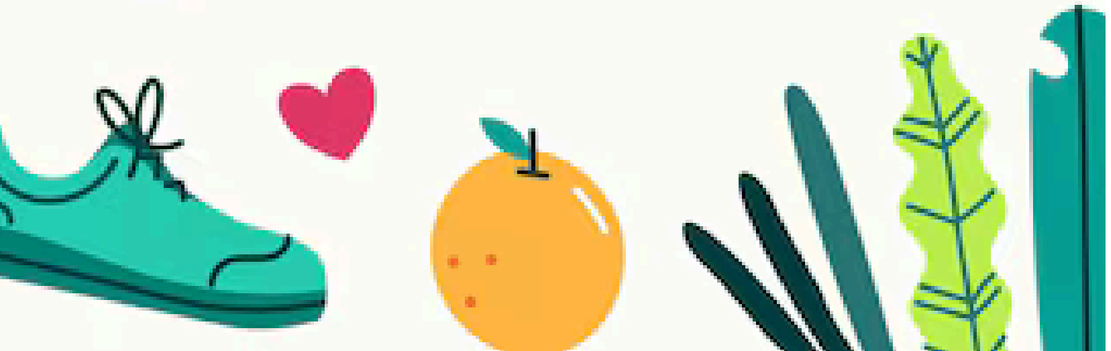
Aegis Life Platform

An Integrated AI-Personalized Health & Wellness Ecosystem



Group ID :R26-IT-136

BSc (Hons) in Information Technology Specialized in Information Technology | March 2026



Small Changes, Big Impact.

Consistency beats intensity—health is a lifestyle, not a sprint.





INTRODUCTION

Aegis Life is an integrated AI health ecosystem where four modules share real-time data to automatically personalize diet, exercise, and wellness for Sri Lankans with complete on device privacy.







PROBLEM BACKGROUND



- Fragmented & Inefficient Dietary Tracking
- Generic Meal Plans Without Health Integration
- High Risk of Exercise Injinjuries Without Proper Guidance
- Disconnection Between Diet and Exercise Goals
- Fragmented Health Data Across Multiple Platforms
- Delayed Detection of Health Issues
- Growing Mental Health Crisis with Limited Support
- No Connection Between Mental and Physical Health





STAKEHOLDERS & IMPACT



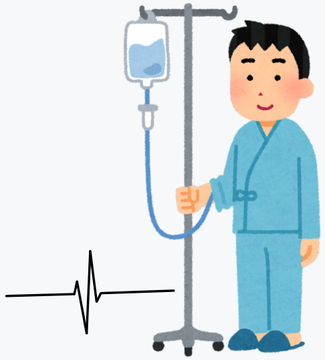
General Users (Public)

Enables easy food tracking through AI scanning leading to better dietary choices



Fitness Beginners

Learns correct exercise form through AI feedback preventing injuries



Chronic Disease Patients

Receives continuous health monitoring and early warning alerts



University Students

Manages academic stress and improves sleep patterns for better performance

EXISTING LIMITATIONS

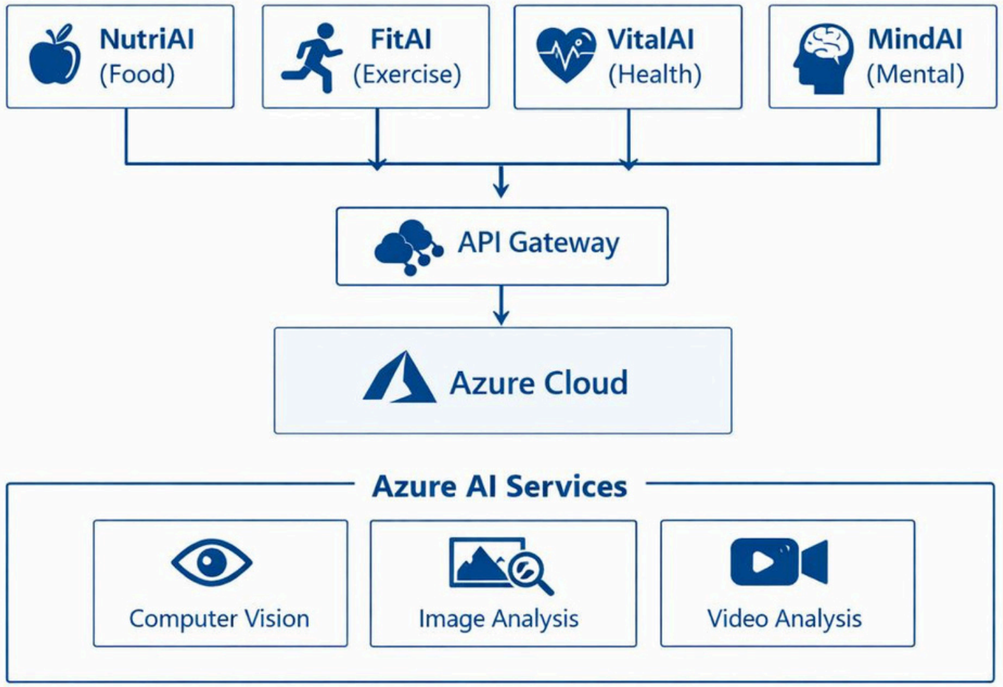
Current Methods

- Fragmented single-purpose health apps (diet, exercise, medical, mental wellness separate)
- Just digital video instructions
- Generic health recommendations not personalized to individual user data
- Basic standalone chatbots with no access to user's complete health profile

The Gap

- No unified platform that integrates diet, exercise, medical, and mental wellness data
- Lack of cross-module intelligence No real-time feedback, just digital video instructions, cannot adapt to health conditions
- No real-time synchronization between health metrics and lifestyle recommendations
- Siloed data prevents holistic, personalized health optimization

SYSTEM ARCHITECTURE



Data Layer

- User Profile Database
- Health Metrics Database
- Nutrition Database
- Exercise Library Database

Processing Layer

- Image Recognition Engine
- Pose Estimation Engine
- NLP Chatbot Engine
- Recommendation Engine

Decision Layer

- Central ML Orchestrator
- Predictive Health Models
- Adaptive Goal Manager
- Smart Alert System

Interface Layer

- Mobile Application with 4 Integrated Modules
- Camera Interfaces
- Chat Interfaces
- Dashboard Visualizations



TEAM CONTRIBUTIONS



Team Member :
W.M.R.A.B.Wijesooriya
Component : FitAI
IT22249098



Team Member :
Dhananjaya A G G
Component : VitalAI
IT22145734



Team Member :GUNATHILAKE
D R G L
Component : MindAI
IT22544636



Team Leader : Saisuhani L
Component : NutriAI
IT22091598



COMPONENT 1: FitAI

IT22249098



W.M.R.A.B.Wijesooriya

Problem Definition

- **High Cost**: Professional trainers cost Rs. 3,000+/hour - unaffordable for most Sri Lankans
- **App Failures**: Current apps don't adapt to health conditions or correct form - causing 52% injuries & 70% abandonment

Research Objectives

- **Dataset Creation**: Create Sri Lankan exercise dataset with 500+ coach videos to train pose estimation model
- **System Development**: Build health-aware fitness app with Sinhala/Tamil feedback that adapts workouts for diabetic patients and achieves 85% accuracy



Novelty

- **Real-Time Form Correction in Local Languages**: Using a trained model that was trained from a dataset that I created and delivers instant voice feedback in Sinhala/Tamil
- **Health-Aware Workout Adaptation-**Automatically adjusts exercise intensity based on the user's health data - gentle workouts for diabetics, BP-safe movements for hypertensive patients - a feature absent in all existing apps.

COMPONENT 1: FitAI

IT22249098



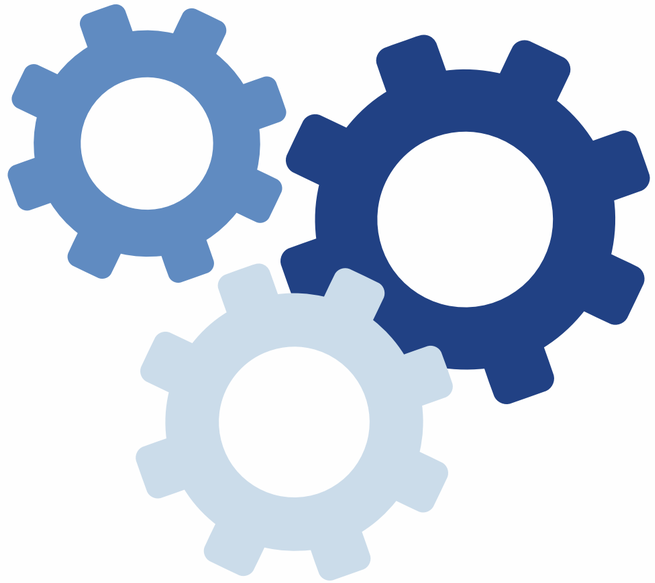
W.M.R.A.B.Wijesooriya

Methodology

- **Research Approach:** Design Science Research with experimental validation using 30 participants over 4 weeks
- **Pose Estimation & Recognition:** MediaPipe tracks 33 body points + TensorFlow Lite identifies 15 exercises in real-time
- **Form Correction & Health Adaptation:** Rule-based ML compares form against 500+ coach videos + automatically adjusts workouts using VitalAI health data
- **Development Tools:** React Native app with on-device processing (MediaPipe, TensorFlow Lite) + Firebase backend
- **Dataset & Ethics:** 500+ coach videos + 50 volunteer videos with informed consent, anonymized data, and SLIIT ethics approval

Technologies

- **React Native** – Cross-platform mobile app
- **MediaPipe** – Real-time 33-point body tracking
- **TensorFlow Lite** – On-device exercise recognition
- **500+ Sri Lankan Videos** – Training dataset
- **Firebase** – Backend + VitalAI integration



COMPONENT 1: FitAI

IT22249098



W.M.R.A.B.Wijesooriya

Validation

- **Pose Estimation Accuracy –** Measures how accurately the system detects user exercise movements.
- **Feedback Latency –** Measures how quickly the system provides real-time feedback.
- **User Satisfaction –** Measures how satisfied users are with the FitAI experience.
- **Health Adaptation –** Ensures exercises safely adapt to users with health conditions.
- **Adherence Improvement –** Measures whether FitAI increases user exercise consistency.

Data Availability

- **Sri Lankan Dataset** – 500+ coach exercise videos and 50 local volunteers representing Sri Lankan body types, and home lighting conditions.
- **Validation Metrics** – Performance tested using ground truth accuracy and user studies .



Security and Ethics

- **Security –** Camera data is processed on-device without cloud upload, and no videos are stored without user consent; all stored data is encrypted.
- **Ethics –** Participants provide informed consent (Sinhala/Tamil), and all collected data is anonymized with access limited to the research team.

COMPONENT 1: FitAI
IT22249098



W.M.R.A.B.Wijesooriya

System Overview

- Camera Input → MediaPipe Pose (33 landmarks) → TensorFlow Lite Exercise Classifier (15 exercises)
- Real-time Form Correction + Sinhala/Tamil Feedback + Health-Adaptive Planner with Firebase integration



- ## Dependencies
- MediaPipe, TensorFlow Lite (Sri Lankan dataset), React Native Expo Camera
- Firebase backend, VitalAI health data,

COMPONENT 2: VitalAi

IT22145734



Dhananjaya A G G

Problem Definition

- 1 in 3 Sri Lankan adults has diabetes or hypertension, yet 90% cannot afford regular checkups or access specialists—early warnings are missed until it's too late
- Existing apps are fragmented, English-only, and reactive. No system offers integrated Sinhala/Tamil health tracking with predictive early warnings

Research Objectives

- **Build AI-Powered Early Warning Algorithms**-To develop AI early warning algorithms that detect rising risk for hypertension and diabetes by analyzing health trends, delivering alerts 2–4 weeks before symptom onset
- **Implement a Multilingual AI Symptom Checker**-To implement a multilingual AI symptom checker in Sinhala and Tamil that provides basic health guidance and recommendations through natural conversation



Novelty

- **Centralized Health Data Integration Hub**: The first AI system to create a Sri Lankan health dataset, train ML models on local data, and predict future disease risk
- **First Sinhala/Tamil AI Symptom Checker-**No existing system offers AI-powered symptom checking in Sri Lanka's local languages

COMPONENT 2: VitalAI

IT22145734



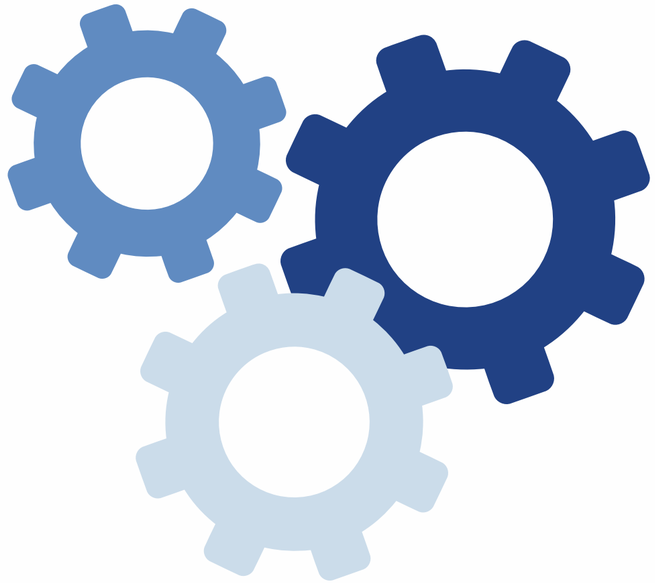
Dhananjaya A G G

Methodology

- **Research Approach:**Design Science Research (DSR) + Quantitative Predictive Modeling
- **Health Data Collection:** AI-powered image scanning using the smartphone camera to extract health indicators from photos.
- **AI Early Warning System:** ML models predict hypertension and diabetes risk 2–4 weeks early
- **Multilingual Symptom Checker:** Dialogflow/Rasa provides Sinhala/Tamil health guidance for 20+ common symptoms
- **Development Tools**: React Native + TensorFlow Lite + Firebase + PostgreSQL + MongoDB
- **Ethics**: SLIIT approval, informed consent, anonymized data
- **Dataset** : Patient health record , Lab reports (blood tests, cholesterol, BMI) ,Estimated 1,000+ records

Technologies

- **React Native** – Front-end
- **TensorFlow Lite** – on-device predictions
- **Dialogflow** – Sinhala/Tamil language support
- **Firebase** – backend integration
- **OpenCV-** Image preprocessing for lab report scanning



COMPONENT 2: VitalAI
IT22145734



Dhananjaya A G G

Validation

- **Predictive Accuracy –** Measures how accurately the system predicts hypertension and diabetes risk 2–4 weeks early
- **Early Warning Lead Time –** Measures how many days in advance the system detects rising health risks before traditional symptom onset
- **Symptom Checker Accuracy –**Measures how accurately the AI identifies and classifies user symptoms in Sinhala and Tamil
- **User Satisfaction –**Measures user satisfaction with VitalAI experience. on System Usability Scale
- **Early Detection Rate–**Measures number of concerning health trends identified requiring medical consultation

Data Availability

- **Hospital & Laboratory Dataset** – 1,000+ anonymized patient records from Sri Lankan hospitals and laboratories for hypertension and diabete
- **Sinhala/Tamil Language Data** – Custom training data using Sri Lankan medical terms, common symptom descriptions, and healthcare conversations in local languages



Security and Ethics

- **Security –** Health data is processed on-device, encrypted in Firebase, and shared only with user consent
- **Ethics –** Ethics approval obtained from the SLIIT Research Ethics Committee, with hospital data collected under official agreements and full patient anonymization.

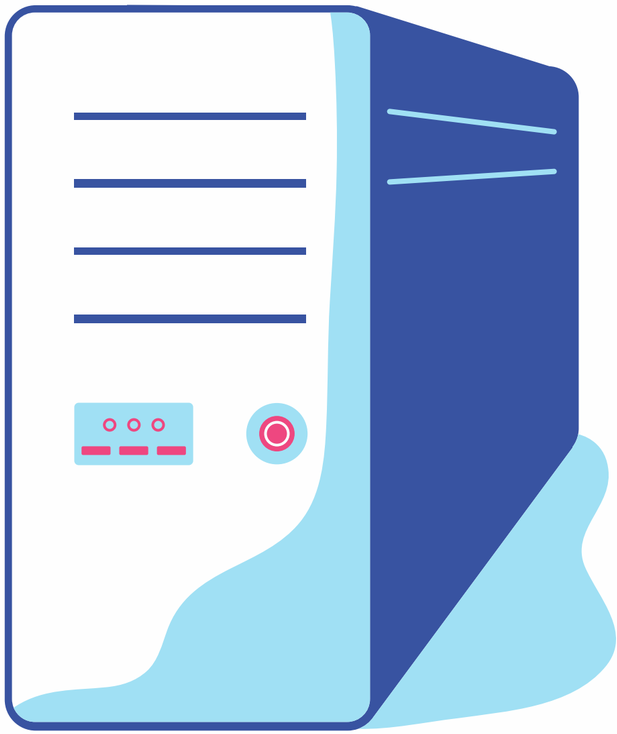
COMPONENT 2: VitalAi
IT22145734



Dhananjaya A G G

System Overview

- Camera Input → Lab Report Scanning
- Real-time Health Alerts → Early Warnings + Triage Recommendations
- MongoDB → Structured Health Records + Flexible Symptom Logs



- Dependencies
- MediaPipe, TensorFlow Lite (Sri Lankan dataset), React Native Expo Camera
- Firebase backend, NutriAI, health data, MongoDB

COMPONENT 3: MindAI

IT22544636



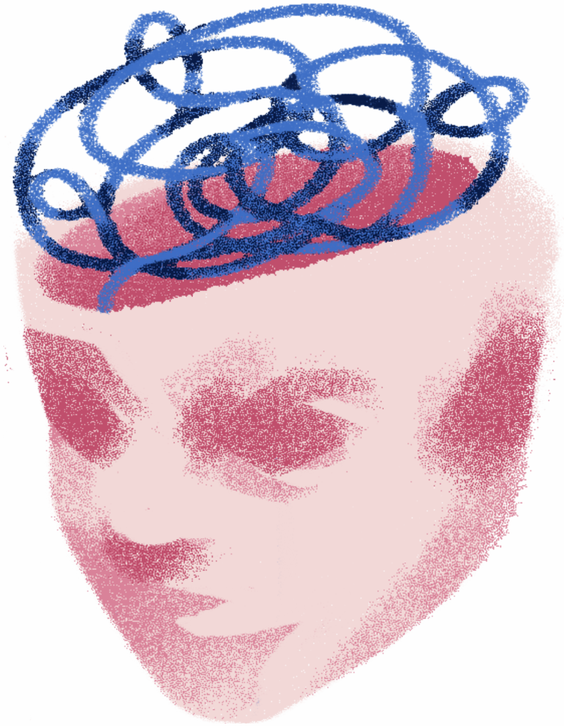
Gunathilake D R G L

Problem Definition

- **High Cost**: Therapy costs Rs. 3,000–8,000 per session | unaffordable for 90% of Sri Lankans
- **No Access**: Only 0.2 psychiatrists per 100,000 people outside Colombo rural areas have zero support
- **App Failures**: Current apps are English-only, culturally disconnected, and offer no 24/7 crisis support – 40% of students suffer in silence

Research Objectives

- **Chatbot Development**: Build 24/7 AI chatbot providing empathetic mental wellness support in Sinhala/Tamil with real-time crisis detection and helpline escalation
- **Content Creation**: Develop culturally adapted meditation library featuring Buddhist practices (Anapanasati, Metta Bhavana) with offline access in local languages
- **Health Integration**: Connect with VitalAI and FitAI to detect early crisis signs from sleep patterns, activity levels, and mood logs for proactive intervention



Novelty

- **First Sinhala/Tamil Mental Health Chatbot**: 24/7 AI powered emotional support with real-time crisis detection no other app offers this in local languages
- **Culturally Adapted Buddhist Meditation**: First app featuring authentic Anapanasati and Metta Bhavana practices with offline access not Western mindfulness imports
- **Face Recognition for Mood Detection**: Real time facial expression analysis to detect stress and anxiety triggering immediate wellness interventions before crisis

COMPONENT 3: MindAI

IT22544636



Gunathilake D R G L

Methodology

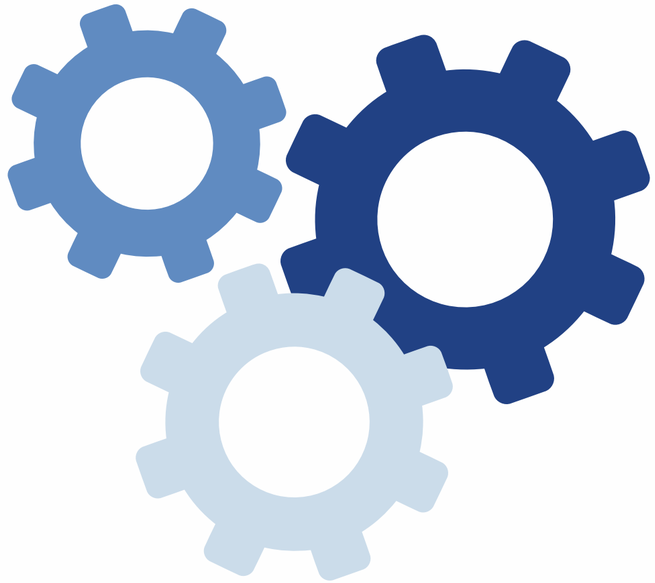
- **Face Recognition:** Implement real time facial expression analysis using TensorFlow Lite to detect stress, anxiety, and mood changes from camera input

- **Chatbot Development**: Build using Rasa/Dialogflow with Sinhala/Tamil NLP training on emotional conversation datasets for empathetic responses

- **Meditation Content**: Record audio with local experts featuring Buddhist practices (Anapanasati, Metta Bhavana) – optimized for offline access

Technologies

- **Mobile Development**: React Native with Expo | cross platform app with camera and audio playback
- **Face Recognition**: TensorFlow Lite + MediaPipe | real-time facial expression analysis and emotion detection
- **Chatbot Framework**: Rasa / Dialogflow | for Sinhala/Tamil NLP and empathetic conversation training
- **Backend & Database**: Firebase | user authentication, mood tracking storage
- **Integration**: RESTful APIs | connect with VitalAI and FitAI for behavioral data sharing



COMPONENT 3: MindAI

IT22544636



Gunathilake D R G L

Validation

- **Chatbot Response Time:** Real-time conversation without frustrating delays | <2 seconds
- **Emotion Detection Accuracy:** Facial expression analysis | ⩾75%
- **Crisis Detection Accuracy:** Correctly identifying distress indicators from conversations
- **Sleep Improvement:** Self reported sleep quality before vs after | 30% better

Data Availability

- **Conversational Data:** 500+ Sinhala/Tamil emotional conversation samples from volunteers and counseling experts
- **Meditation Content:** 25+ guided audio recordings with Buddhist teachers (Anapanasati, Metta Bhavana) in Sinhala/Tamil
- **Validation Data:** Mood logs, sleep patterns, and PHQ-9/GAD-7 scores from 50+ beta testers



Security and Ethics

- **Security –** End-to-end encryption for all conversations and mood data with anonymized user identities
- **Ethics –** Informed consent with clear crisis protocols – chatbot provides helpline numbers immediately if suicidal ideation detected, and user data is never sold

COMPONENT 3: MindAI
IT22544636

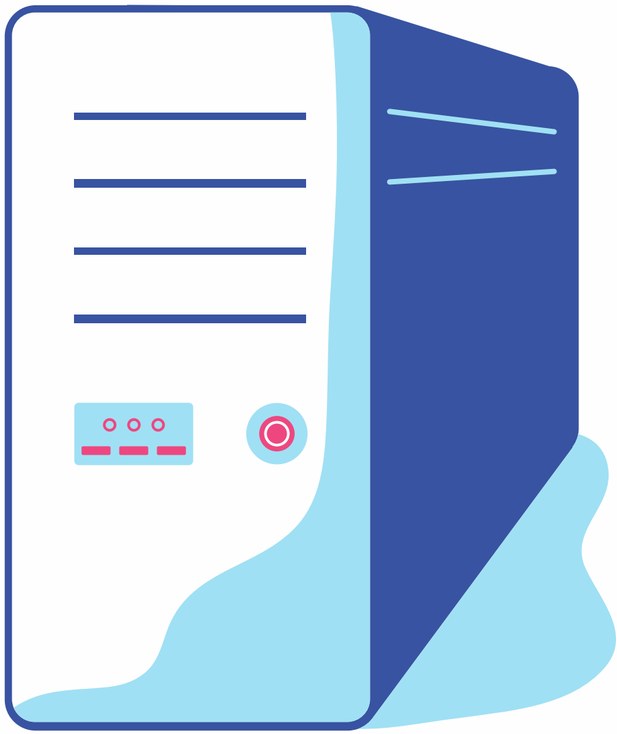


Gunathilake D R G L

System Overview

- **Input Processing:** User messages, mood logs, and sleep data → Rasa NLP engine analyzes Sinhala/Tamil text → Sentiment detection + emotion classification

- **Response & Integration**: Empathetic chatbot replies + meditation recommendations → Crisis detection escalates to helplines → Firebase sync with VitalAI/FitAI for behavioral insights



Dependencies

- MediaPipe, TensorFlow Lite (Sri Lankan dataset), React Native Expo Camera
- Firebase backend, VitalAI health data, FitAI health data

COMPONENT 4: NutriAI
IT22091598



Saisuhani L

Problem Definition

- **Manual Food Logging –** Users must enter meals manually, which is slow and often inaccurate.
- **Global Food Databases** – Most nutrition apps focus on Western foods and lack Sri Lankan dishes.
- **Limited Personalization** – Many systems give generic diet plans without considering local foods or budget.

Research Objectives

- **Sri Lankan Food Dataset:** Create a dataset with 1,000+ labeled images of traditional Sri Lankan foods.
- **Nutrition Database:** Localized Nutrition Database – Build a database with nutrition and cost data for 200+ Sri Lankan foods.



Novelty

- **Localized Food Recognition –** Custom dataset for Sri Lankan foods such as rice & curry, hoppers, and kottu.
- **Budget-Aware Recommendation –** Meal suggestions based on nutrition needs and user budget.

COMPONENT 4: NutriAI
IT22091598



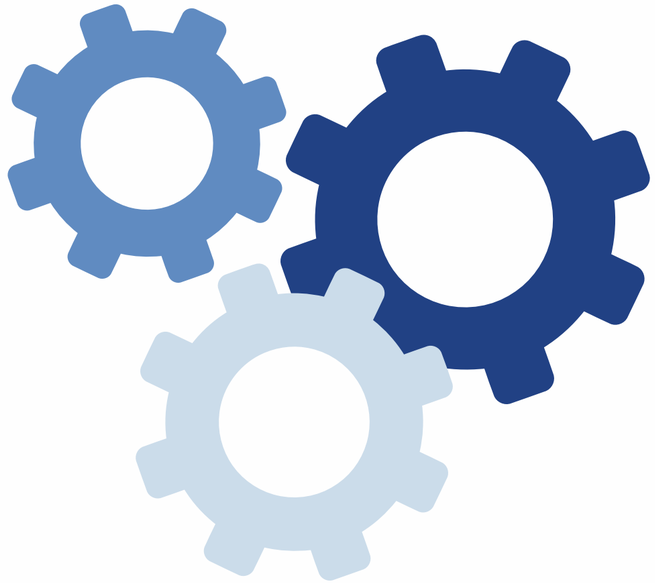
Saisuhani L

Methodology

- **Research Approach:** Design Science Research with experimental validation using 500+ meal samples.
- **Food Recognition:** Use CNN models in TensorFlow to classify Sri Lankan food images.
- **Budget-Aware Algorithm:** Recommend meals that meet nutrition targets at minimum cost.
- **Dataset Collection:** 1,000+ Sri Lankan food images; Sources: restaurant meals, home cooking, online images; Ethics approval from SLIIT Research Ethics Committee.

Technologies

- **React Native** – Mobile app development
- **OpenCV** – Image preprocessing
- **TensorFlow Lite** – On-device food recognition
- **Custom Dataset –** Sri Lankan food images
- **Firebase** – Backend and data storage



COMPONENT 4: NutriAI
IT22091598



Saisuhani L

Validation

- **Food Recognition Accuracy –** Correct identification of Sri Lankan foods.
- **Multi-Item Detection Rate –** Ability to detect multiple foods in one meal.
- **Portion Estimation Error –** Accuracy of portion size estimation.
- **Nutrient Calculation Accuracy –** Comparison with dietitian analysis.
- **Budget Compliance Rate-** Measures if recommendations stay within user budget.

Data Availability

- **Sri Lankan Food Image Dataset** – 1,000+ labeled food images
- **Sri Lankan Nutrition Database**
- **Sri Lankan Food Price Database**

Security and Ethics

- **Security –** Food recognition runs locally on device; Images shared only with user consent; Data protected using Firebase encryption.
- **Ethics –** Approval from SLIIT Research Ethics Committee.



COMPONENT 4: NutriAI
IT22091598



Saisuhani L

System Overview

NutriAI is an AI-based Sri Lankan food recognition and nutrition management module in the Aegis Life platform.

Users can:

- Capture food images using the camera
- Upload meal images from the gallery
- Automatically detect Sri Lankan foods
- View nutrition analysis and budget-aware recommendations



VALIDATION & ETHICS



Validation

90%+ target for Accuracy, Precision, Recall, exercise recognition and form correction



Privacy

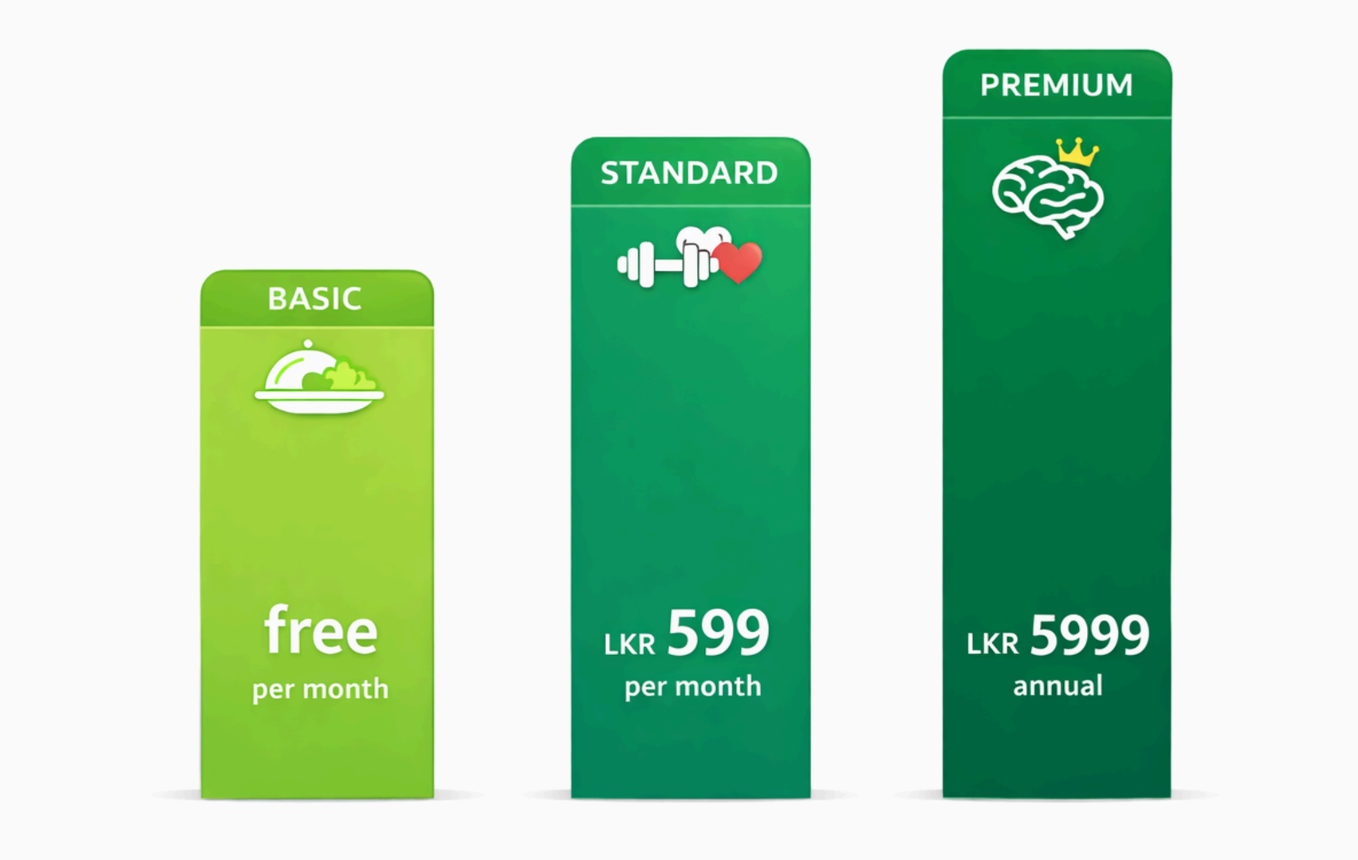
Real-time on-device pose extraction with instant frame deletion - no exercise videos stored or uploaded



Security

End-to-end encryption + strict access control - only authenticated users access their own health data

MARKET & COMMERCIALIZATION

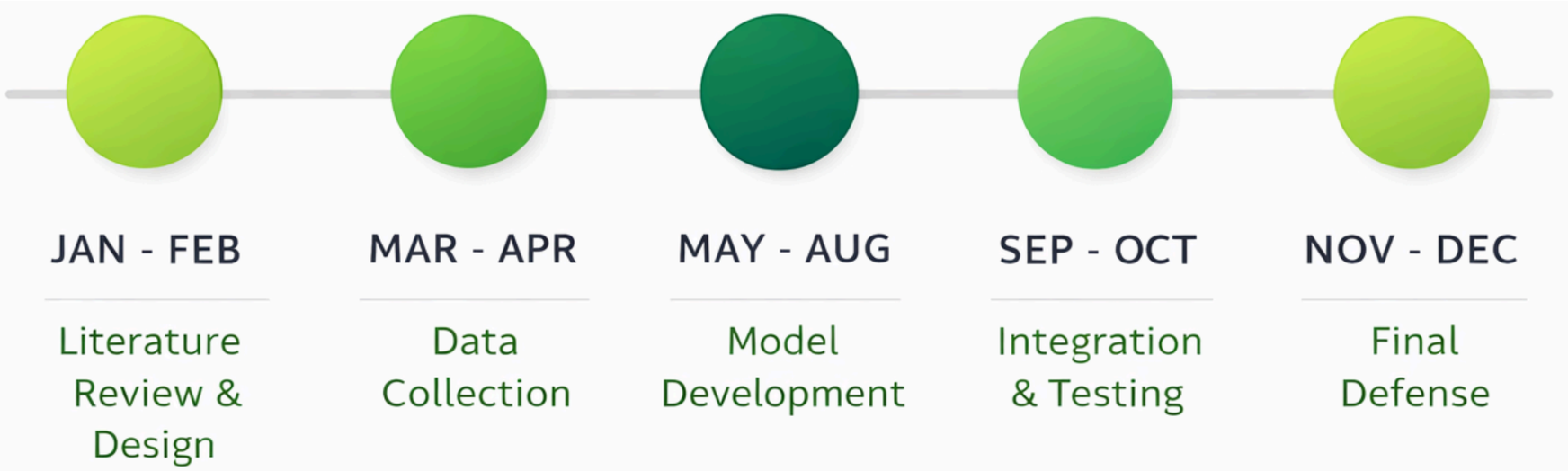


- Health-Conscious General Public
- Chronic Disease Patients (Diabetes, Hypertension)
- Fitness Enthusiasts
- Corporate Wellness Programs





TIMELINE & WBS



WBS: Project Lead → Research → ML Training → Hardware Integration → QA Testing



Thank You

Thank you for your attention
We welcome any questions or feedback from the panel.